

THE DISTILLER – 40 min

Objective of Experiment No. 11

To learn how to build and use a liquid distiller.

Equipment

- 1 × Bunsen burner
 - 1 × Tripod
 - 1 × Wire gauze
 - 3 × Pieces of masking tape (not included)
 - 1 × Plastic clamp (not included)
 - 1 × Latex glove (not included)
 - 1 × 100 ml beaker
 - 1 × 250 ml beaker
 - 50 ml tap water
 - 50 ml alcohol (not included)
 - 1 × Small rubber stopper (13 × 5 × 17 mm)
 - 1 × Alcohol thermometer
 - 2 × Ice cubes
 - 1 × U-shaped glass drying tube
 - 2 × Clamps
 - 1 × Scissors (not included)
 - 1 × Silicone tube (30 cm)
 - 1 × Plastic base
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Description of the experiment

- Using scissors, cut off the four upper fingers of the latex glove, leaving the thumb intact. Similarly, cut a 30 cm piece of silicone tubing (see STEP 1).

- Take the modified glove and place it over the larger 250 ml beaker, securing it with masking tape to ensure a tight seal. Then insert the silicone tube into the glove and fix it with a plastic clamp (see STEP 2).
- Insert the other free end of the silicone tube into the U-shaped drying tube, securing it again with masking tape (see STEP 3).
- Position the clamps and plastic base to support the smaller 100 ml beaker. Place the U-shaped drying tube inside it and close its free end with the rubber stopper (see STEP 4).
- Return to the larger 250 ml beaker and cut the tip of the glove's thumb. Use this opening to pour 50 ml of tap water and 50 ml of alcohol into the beaker. Then insert the thermometer through the same opening and seal it with masking tape.
- Place the two ice cubes inside the smaller 100 ml beaker to cool the U-shaped drying tube as much as possible (see STEP 5). Then light the Bunsen burner and wait until the temperature of the solution (water + alcohol) reaches about 65°C. At this point, the alcohol will begin to evaporate and travel through the silicone tube to the cooled drying tube in the other beaker.
- Observe the condensation of alcohol on the cold surface of the drying tube.

Final observations

Distillation is a technique used to separate two or more substances in a mixture.

From ancient industries to modern ones, this technique has been widely used for producing distilled products obtained by separating alcohol from heterogeneous mixtures derived from processing cereals or fruits.

Similarly, in this experiment, the goal is to extract the alcohol contained in a heterogeneous mixture of water and alcohol. The distillation process exploits the significant difference in boiling temperatures between alcohol (65°C) and water (100°C).

Once the temperature reaches 65°C, the alcohol in the mixture begins to evaporate and flows through the silicone tube until it reaches the U-shaped drying tube, where it condenses due to the cold surface created by the ice in the beaker.

During the entire alcohol extraction process, the temperature measured by the thermometer remains constant. An increase in temperature indicates that the alcohol in the mixture has been completely evaporated.